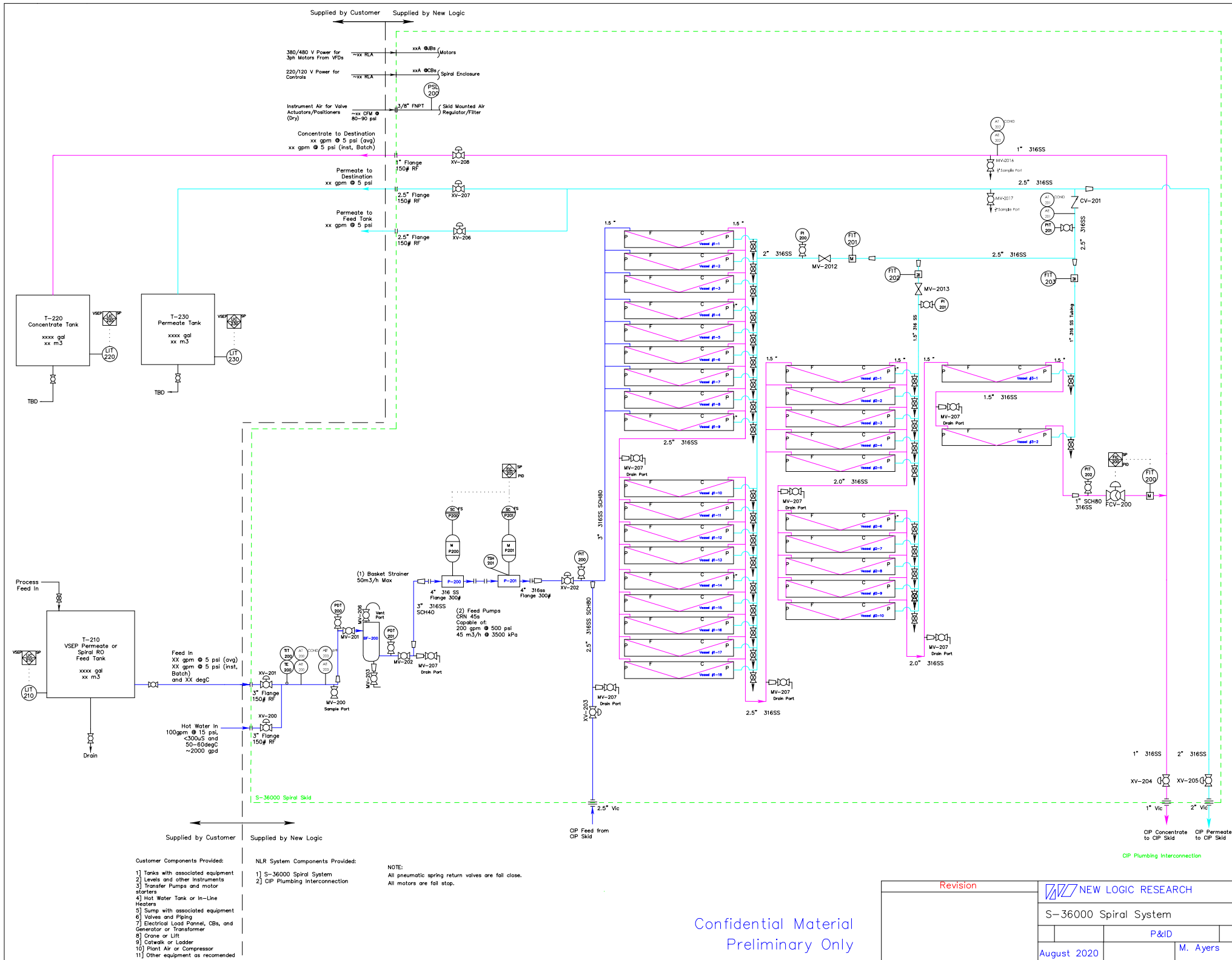




4

3

2

1

## ISO FRONT VIEW

Pumps and Filters  
S-36000-04  
Spiral Housing (30x)  
PRO-8-600-SP-3

Feed Plumbing  
S-36000-02-03

Permeate Plumbing  
SA-36000-02-01  
SA-36000-02-01.1  
SA-36000-02-01.2  
SA-36000-02-01.3

Pump CRN 45-6 A-P-GI-E-HQQE  
Pump CRN 45-7 SF SF-P-GI-E-HQQE

Concentrate Pumping  
S-36000-02-02

Bag Filter with 100 mesh

Feed Inlet Plumbing  
S-36000-02-300

Electrical Components  
S-36000-05  
Enclosure A36H30CLP

Permeate to CIP Tank

Concentrate to CIP Tank

Feed from CIP Skid

Frame  
S-36000-01

## NOTES:

1. New Logic Research confidential material.
2. Slight differences between drawings and actual system may be attributed to New Logic Research Continuously evolving and improving its technology.
3. All dimensions are shown in inches [mm] and for reference only.

## S-36000 Spiral System

# ISO BACK VIEW

Concentrate to Destination

Feed In

Hot Water In

Permeate to Destination

Permeate to Feed Tank

Permeate Plumbing  
S-36000-02-01

Permeate Plumbing  
S-36000-02-01.1

Permeate Plumbing  
S-36000-02-01.2

Permeate Plumbing  
S-36000-02-01.3

## NOTES:

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## S-36000 Spiral System

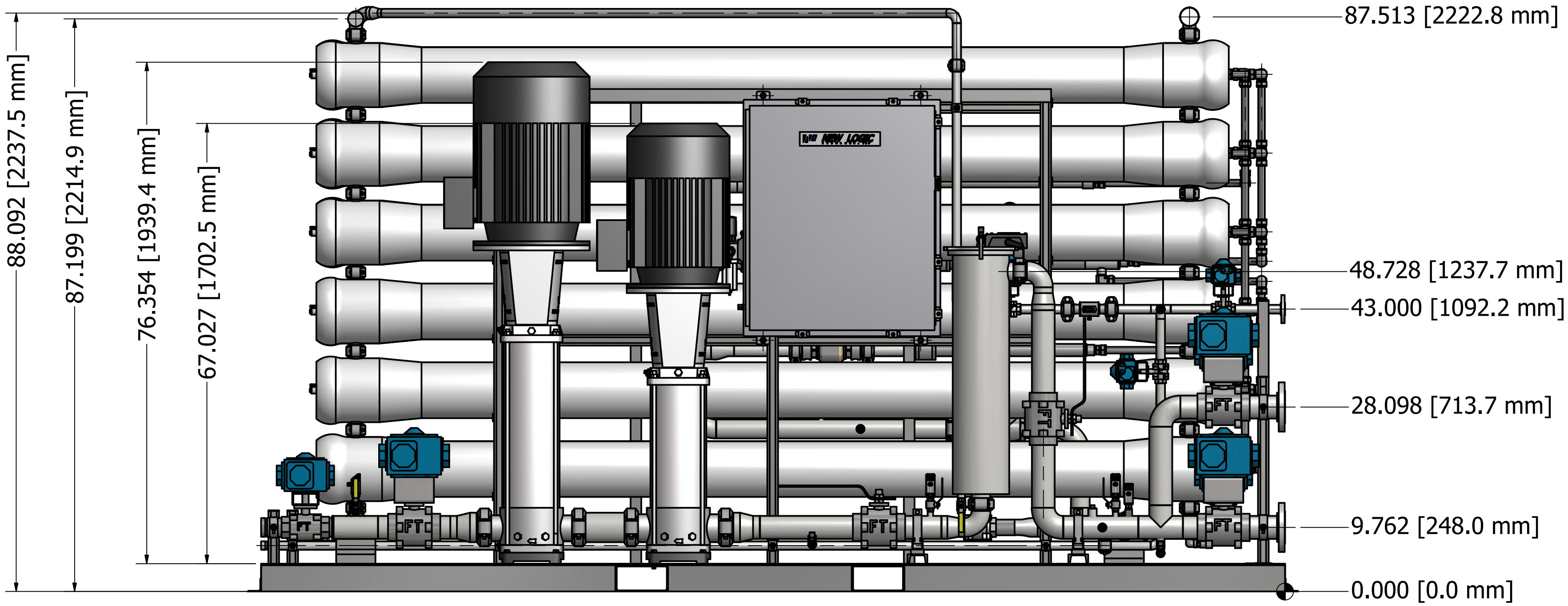
4

3

2

1

## FRONT VIEW



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## S-36000 Spiral System

4

3

2

1

1

## BACK VIEW

D

D

87.513 [2222.8 mm]

78.759 [2000.5 mm]

66.759 [1695.7 mm]

62.204 [1580.0 mm]

C

C

54.759 [1390.9 mm]

50.283 [1277.2 mm]

47.236 [1199.8 mm]

42.759 [1086.1 mm]

36.764 [933.8 mm]

30.759 [781.3 mm]

B

B

18.759 [476.5 mm]

4.250 [108.0 mm]

4.000 [101.6 mm]

0.000 [0.0 mm]

B

B

8.000 [203.2 mm]

36.000 [914.4 mm]

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S-36000 Spiral System

4

3

2

1

1

A

A

4

3

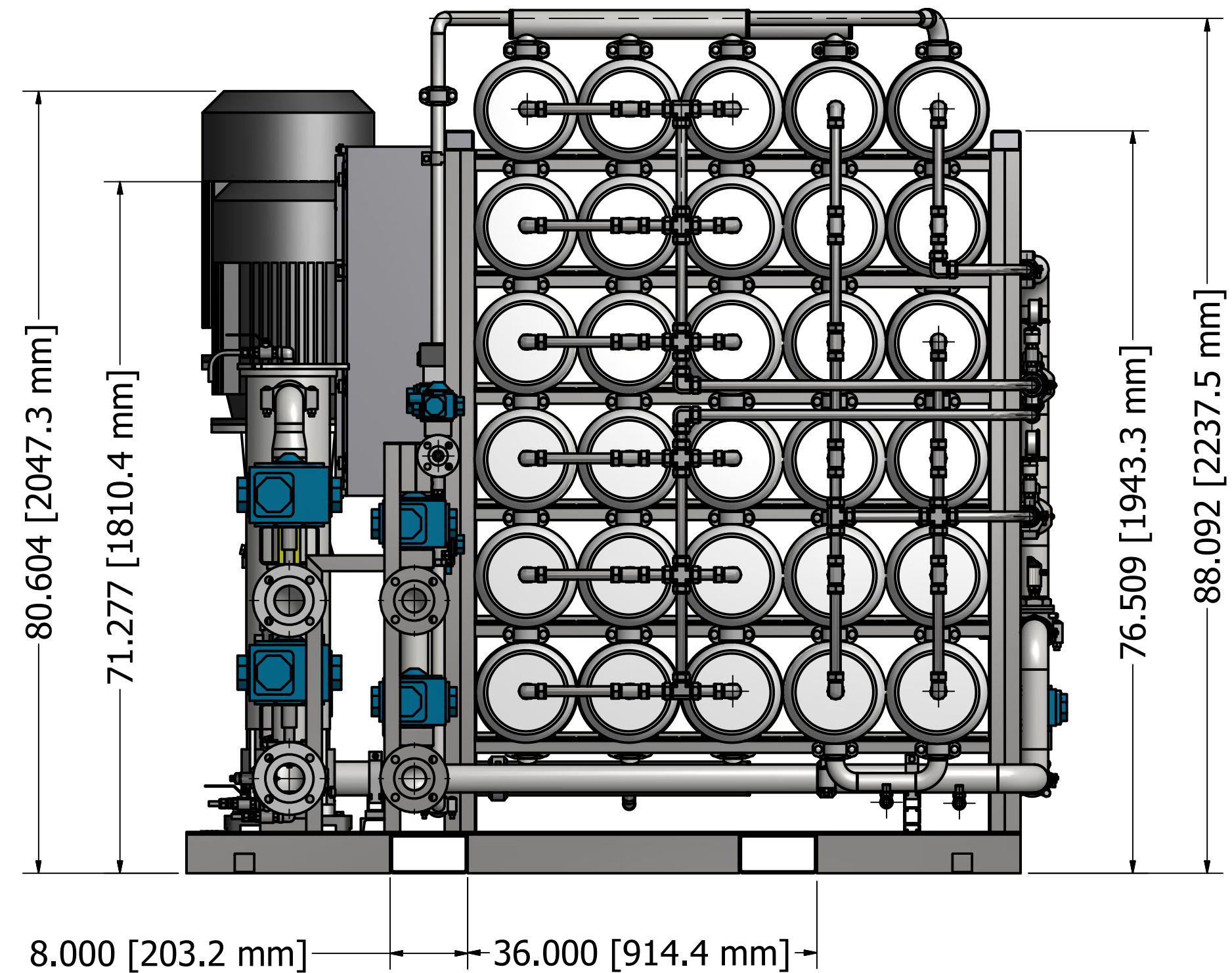
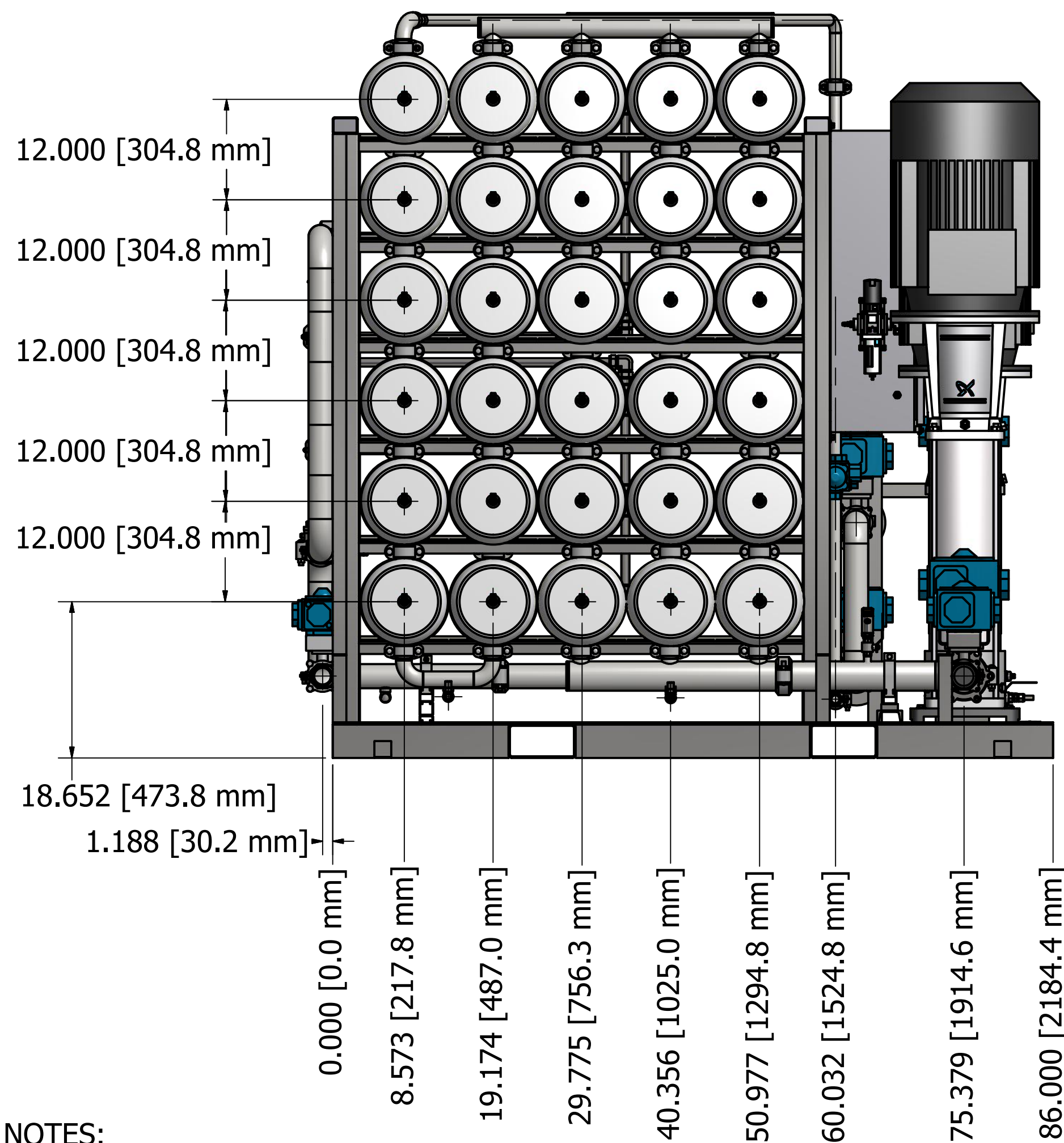
2

1

## SIDE VIEWS

LEFT

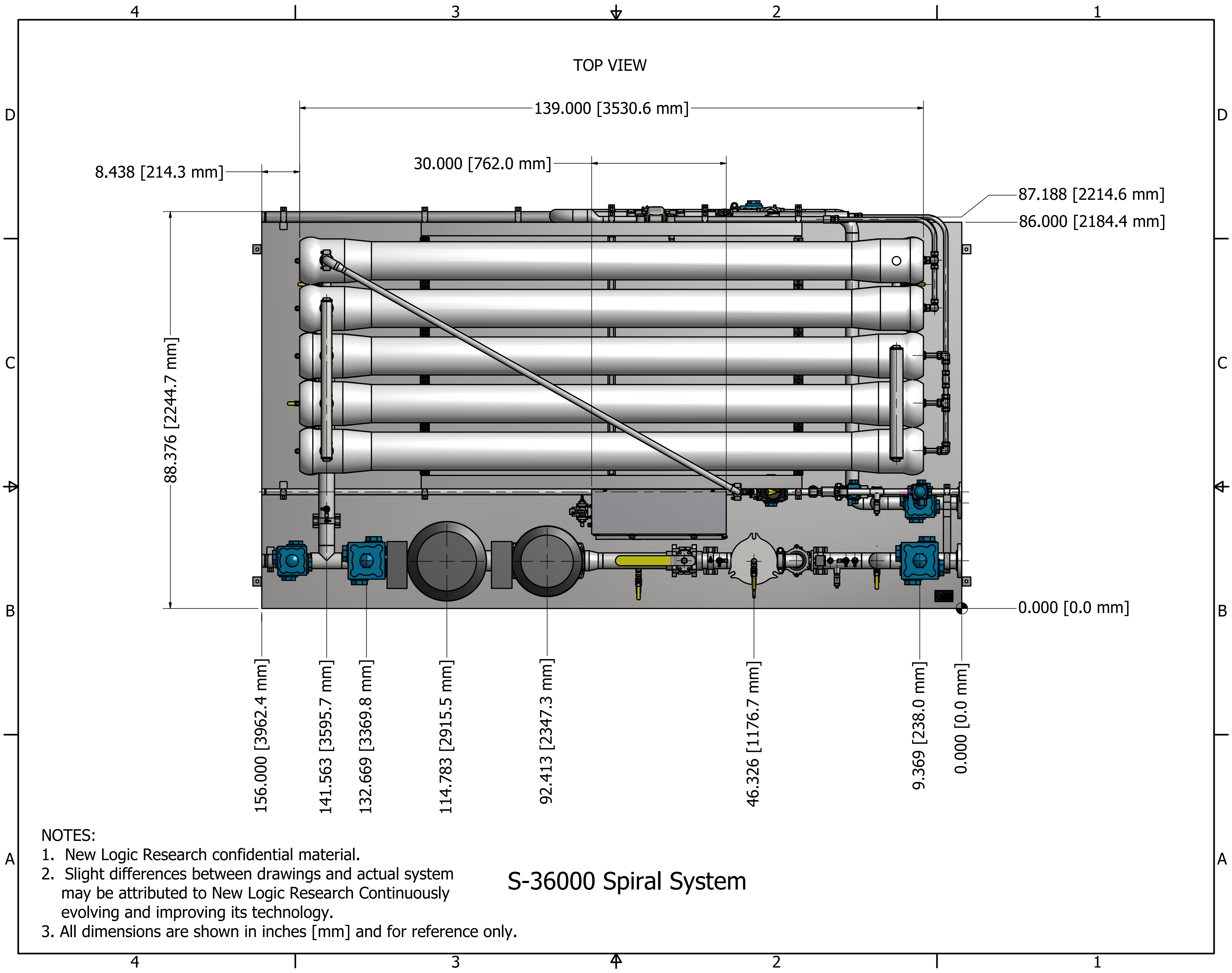
RIGHT



## NOTES:

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S-36000 Spiral System



## SPIRAL SYSTEM COMPONENT SPECIFICATIONS

### S-36000 Spiral System

#### 1] Filter Pack

|                                 |                                                           |
|---------------------------------|-----------------------------------------------------------|
| Membrane:                       | MF, UF, NF, RO (depends on application and pilot testing) |
| Membrane Surface Area:          | 36000sf (3344m <sup>2</sup> ) (90ea 8"Dia x 40"L)         |
| Estimated Production:           | 125-250gpm (at 5-10GFD)                                   |
| Maximum Operating Temperature:  | 120 °F (49°C) (Options for up to 80°C)                    |
| Allowable Feed Slurry pH Range: | 2-11 (Options for 1-14)                                   |
| O-ring Material:                | Ethylene Propylene                                        |
| Filter Pack Housing:            | Polyurethane Coated, Filament Wound, Epoxy FRP            |
| Wetted Ports:                   | 316L Stainless Steel                                      |
| Wetted Plastic End Plates:      | PVC Thermoplastic                                         |

#### 2] Piping

|                         |                                      |
|-------------------------|--------------------------------------|
| Maximum Pressure:       | 550 psi (Options for up to 1200 psi) |
| Pressure Piping:        | 316L Stainless Steel Schedule 80     |
| Atmospheric Piping:     | 316L Stainless Steel Schedule 40     |
| Pipe Clamps:            | Zero-Flex Victaulic Style 07 EPDM    |
| On/Off Valves:          | 316 SS 3-piece ball valve            |
| Flow Control Valves:    | 316 SS V-port ball valve             |
| Check Valves:           | 316 SS swing check valve             |
| Pneumatic Actuators:    | Alpha Compact 4, Spring Return       |
| Positioners:            | BLX Linear Quarter Turn I/P          |
| Skid Pipe Terminations: | 150 lb 316 SS Raised Face Flange     |
| Flange Gaskets:         | 150 lb Garlock Blue Guard            |

#### 3] Pump Specifications:

|                         |                                                 |
|-------------------------|-------------------------------------------------|
| Feed Pump Type:         | Grundfos 316SS Vertical Multi-stage Centrifugal |
| Feed Pump Motor:        | Grundfos ML 3500 rpm TEFC 460VAC, 3phase        |
| Motor Speed Controller: | Danfoss 380-480VAC, NEMA 4 (or Equal)           |

#### 4] Pre-Screen Basket Strainer:

|                      |                                             |
|----------------------|---------------------------------------------|
| Filter Housing Type: | NLR Custom Made 316 SS                      |
| Basket Strainer:     | 100 Mesh 316 SS (Optional Felt Bag Filters) |
| Capacity:            | 120 GPM, 4.6 sqft                           |
| Elastomers (Seals):  | EPDM (option for Viton)                     |

#### 5] Operating Site Conditions:

|                        |                                                      |
|------------------------|------------------------------------------------------|
| Equipment Rating:      | NEMA 4, Indoor/Outdoor protect from sunlight & rain. |
| Ambient Temperature:   | 5 - 37°C                                             |
| Storage Temperature:   | 2 - 70°C (Protect Filter Pack from Freezing)         |
| Max Relative Humidity: | <95%, non-condensing                                 |
| Elevation:             | 3300 ft maximum without derating                     |

**SPIRAL SYSTEM COMPONENT SPECIFICATIONS (continued)****6] Electrical Specifications:**

(Note typically this is provided with a VSEP system, and the PLC/HMI is split with the spiral system. The controls are mounted on CIP 'Clean in Place' Skid that is also shared)

|                                 |                                                      |
|---------------------------------|------------------------------------------------------|
| Controls Voltage:               | 120/220VAC, 1phase, 15A, 50/60Hz                     |
| Programmable Controller Type:   | None (Control signals from CIP PLC)                  |
| Touch Screen Display Type:      | None (Set Point change at CIP HMI)                   |
| Enclosures:                     | <i>Hoffman</i> NEMA 4 Powdercoat Finish 20x16x8      |
| Switches, Relays, & Contactors: | Allen Bradley 120/220 Volt                           |
| IO Link Masters and Hubs:       | IFM Efector AL1122                                   |
| Pressure Sensors:               | IFM Efector PN-2292                                  |
| Mag Flow Tube and Transmitter:  | IFM Efector SM2000, SM8000                           |
| Bag Filter DP Sensors:          | IFM Efector PN-2294                                  |
| Temperature Sensor:             | IFM Efector TN-2511, UT0028                          |
| Conductivity Meter:             | IFM Efector LDL-100                                  |
| Solenoid Valves:                | Festo Manifold Gang Mounted Dual 3/2 24 VDC, IO Link |
| VSEP and CIP Communication:     | IO Link                                              |
| VFD Communication:              | Ethernet Com Card                                    |
| DCS Communication:              | Ethernet OPC Server (option for Modbus)              |

|                                                                                                                                                                                                                                                                                                        |               |                            |                      |                   |                       |  |                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------|----------------------|-------------------|-----------------------|--|--------------------|
| Utility Summary                                                                                                                                                                                                                                                                                        |               |                            |                      |                   |                       |  | New Logic Research |
| VSEP System                                                                                                                                                                                                                                                                                            |               |                            |                      |                   |                       |  |                    |
| <b>AIR COMPRESSOR SIZING INFORMATION</b>                                                                                                                                                                                                                                                               |               |                            |                      |                   |                       |  |                    |
|                                                                                                                                                                                                                                                                                                        | # of Valves   | In <sup>3</sup> Air/stroke | Strokes/day          | CFM               | M3/hr                 |  |                    |
| <b>Spiral RO</b>                                                                                                                                                                                                                                                                                       |               |                            |                      |                   |                       |  |                    |
| 1" Pneumatic Valves                                                                                                                                                                                                                                                                                    | 2             | 35                         | 20                   | 0.0006            | 0.001                 |  |                    |
| 2" Pneumatic Valves                                                                                                                                                                                                                                                                                    | 4             | 60                         | 20                   | 0.0019            | 0.0033                |  |                    |
| 3" Pneumatic Valves                                                                                                                                                                                                                                                                                    | 2             | 93                         | 20                   | 0.0015            | 0.0025                |  |                    |
| 1" Flow Control Valves                                                                                                                                                                                                                                                                                 | 1             | 35                         | 20                   | 0.5003            | 0.8505                |  |                    |
| <b>System Air Totals</b>                                                                                                                                                                                                                                                                               |               |                            | <b>System Totals</b> | <b>0.50</b>       | <b>0.8573</b>         |  |                    |
| Supply instument grade dry air at 80-90 psi (560-630kPa)                                                                                                                                                                                                                                               |               |                            |                      |                   |                       |  |                    |
| Supply air to enclosure mounted regulator/filter 3/8" NPT connection                                                                                                                                                                                                                                   |               |                            |                      |                   |                       |  |                    |
| Totals do not include air for tools or other sources.                                                                                                                                                                                                                                                  |               |                            |                      |                   |                       |  |                    |
| <b>HOT WATER SIZING INFORMATION</b>                                                                                                                                                                                                                                                                    |               |                            |                      |                   |                       |  |                    |
|                                                                                                                                                                                                                                                                                                        | # /Day        | Temp degC                  | Gallons/Day          | GPM               | M3/hr                 |  |                    |
| <b>Spiral RO</b>                                                                                                                                                                                                                                                                                       |               |                            |                      |                   |                       |  |                    |
| Cleanings                                                                                                                                                                                                                                                                                              | 0.14          | 45-50                      | 280                  | 0.19              | 0.0441                |  |                    |
| Rinse filter pack at 120gpm for 5mins                                                                                                                                                                                                                                                                  | 0.14          | 45-50                      | 84                   | 0.06              | 0.0132                |  |                    |
| Intermittent need of additional cleaning or flush of filter pack                                                                                                                                                                                                                                       | 0.033         | 45-50                      | 19.8                 | 0.01              | 0.0031                |  |                    |
| <b>System Water Totals</b>                                                                                                                                                                                                                                                                             |               |                            | <b>System Totals</b> | <b>0.27</b>       | <b>0.0605</b>         |  |                    |
| Supply chlorine free water at >300 uS/cm                                                                                                                                                                                                                                                               |               |                            |                      |                   |                       |  |                    |
| Supply water at 45degC and 120gpm (27.3m3/hr)                                                                                                                                                                                                                                                          |               |                            |                      |                   |                       |  |                    |
| Supply water at 20 psi to 2" Hot Water flange connection on Spiral Skid                                                                                                                                                                                                                                |               |                            |                      |                   |                       |  |                    |
| <b>ELECTRICAL LOAD PANEL SIZING INFORMATION</b>                                                                                                                                                                                                                                                        |               |                            |                      |                   |                       |  |                    |
| Circuit Breaker Sizing 1.25-2x Nameplate Current                                                                                                                                                                                                                                                       |               |                            |                      |                   |                       |  |                    |
| 3-Phase Power Voltage VAC                                                                                                                                                                                                                                                                              | 400-460       | 50/60 Hz                   |                      |                   |                       |  |                    |
| 1-Phase Power Voltage VAC                                                                                                                                                                                                                                                                              | 220-120       | 50/60 Hz                   |                      |                   |                       |  |                    |
|                                                                                                                                                                                                                                                                                                        |               |                            |                      |                   |                       |  |                    |
|                                                                                                                                                                                                                                                                                                        | # Motors      | Rated Motor HP             | Nameplate Current    | Nameplate Current | Recommended CB Sizing |  |                    |
|                                                                                                                                                                                                                                                                                                        |               |                            | 400V                 | 460V              |                       |  |                    |
| Spiral RO Feed Pumps                                                                                                                                                                                                                                                                                   | 1             | 50                         | 64.1                 | 56.0              | 100                   |  |                    |
| Spiral RO Feed Pumps                                                                                                                                                                                                                                                                                   | 1             | 60                         | 78                   | 65.0              | 120                   |  |                    |
| <b>Totals</b>                                                                                                                                                                                                                                                                                          | <b>2</b>      |                            |                      |                   |                       |  |                    |
| Supply 3-Phase power from customer supplied load panel (with circuit breakers, disconnects, line filters, etc.) to the inlet of each VFD provided by New Logic.                                                                                                                                        |               |                            |                      |                   |                       |  |                    |
|                                                                                                                                                                                                                                                                                                        | # Connections | Rated Motor                | Nameplate            | Nameplate         | Recommended           |  |                    |
|                                                                                                                                                                                                                                                                                                        |               |                            | 220V                 | 120V              |                       |  |                    |
| Main Control Enclosures                                                                                                                                                                                                                                                                                | 1             | NA                         | 10                   | 8.0               | 20                    |  |                    |
| Supply 1-phase power from customer supplied load panel (with circuit breakers, ups, etc.) to New Logic provided circuit breakers inside the Spiral RO enclosure.                                                                                                                                       |               |                            |                      |                   |                       |  |                    |
| Note: These are estimates only based on very preliminary data. These calculations are subject to change and do not include equipment offskid of VSEP system (ie: Transfer Pumps, Generators, Heaters, Lighting, Cranes, etc). For power consumption estimates refer to the operating cost spreadsheet. |               |                            |                      |                   |                       |  |                    |